PetPal_Ai.git

SYSTEM PURPOSE: PetPal_Ai is a system designed to provide a user-friendly interface for interacting with a pet. It allows users to monitor their pet's status, receive alerts, and access various features such as feeding schedules, playtime, and health tracking. The system aims to provide a comprehensive solution for pet owners to manage their pets' needs and well-being.

TECH STACK: - backend: Python - frontend: Not implemented - database: Not implemented - libraries: Not implemented

MODULES: AI Client: The AI Client module is responsible for handling user interactions, processing user input, and sending requests to the database. It also receives data from the database and updates the user interface accordingly. Files: ai_client.py

Database: The Database module is responsible for storing and retrieving data related to the pet's status, feeding schedules, playtime, and health tracking. It also handles user authentication and authorization. Files: db.py, sqlite_main_app.py

Merge Databases: The Merge Databases module is responsible for merging data from multiple sources into a single database. It also handles data migration and synchronization. Files: merge_databases.py

API ROUTES: - POST /login: This endpoint handles user login requests, authenticating the user and returning a token for subsequent requests. - GET /pet/status: This endpoint returns the current status of the pet, including feeding schedules, playtime, and health tracking information. - POST /pet/feed: This endpoint updates the pet's feeding schedule and sends a notification to the user.

DATA FLOW: - The system starts with user input, which is processed by the AI Client module and sent to the Database module for storage and retrieval. - The Database module retrieves the relevant data and sends it back to the AI Client module, which updates the user interface accordingly. - The user can also interact with the system through API routes, such as logging in, checking the pet's status, and feeding the pet.

ARCHITECTURE: The system architecture is a client-server architecture, with the AI Client module acting as the client and the Database module acting as the server. The system uses a single database to store all data related to the pet's status, feeding schedules, playtime, and health tracking. The Merge Databases module is used to merge data from multiple sources into a single database.

SECURITY: - Not implemented in code

IMPROVEMENTS: - Implement security measures, such as encryption and authentication, to protect user data and prevent unauthorized access. - Optimize the system's performance by using caching and indexing to improve data retrieval and storage. - Add more features, such as reminders and notifications, to enhance the user experience.